STA2002 Probability and Statistics II
Assignment 6
Due date: **December 13th 2023 11:59pm**
Please submit your written answers as a pdf file.

Please work on the assignment by yourself only. We follow a strict rule on academic honesty.

1. (ANOVA1) In this question, we will continue our investigation on the traffic volume dataset. Download the traffic dataset "traffic.csv" from Assignment 2 Blackboard folder. In particular, we would like to investigate the effect of weather conditions on the traffic volume. We shall only look at the effect of three weather: "Clouds", "Clear" and "Snow". These three weather conditions are the "treatments" in ANOVA. For this question, feel free to use any ANOVA function or package in your choice of programming language, and attach your code at the end of your assignment.

   (a) Fill in the following ANOVA table. If you decide to do it in Excel, it is really a few clicks away from the answer.

	SS	df	MS	F
Treatment				
Error				
Total				

   (b) Test the hypothesis that the hourly traffic has the same mean under these 3 weather conditions at $\alpha = 5\%$, assuming that the hourly traffic is normally distributed. State clearly (i) the degrees of freedom that you use in this F-test and (ii) the conclusion of the test.

2. (ANOVA2) We would like to compare the lifetime of four brands of pens. It is suggested that the writing surface might affect lifetime, so the following lifetimes (in minutes) are collected for each brand-surface combination:

		Writing Surface		
		1	**2**	**3**
Brand of Pen	**1**	709, 659	713, 726	660, 645
	2	668, 685	722, 740	692, 720
	3	698, 685	666, 684	678, 750
	4	698, 650	704, 666	686, 733

(a) Fill in the following ANOVA table:

	SS	df	MS	F
Row				
Column				
Interaction				
Error				
Total				

(b) Is there an effect on the lifetime of pen due to the brand (i.e. row effect)? Conduct the appropriate test at $\alpha = 5\%$.

(c) Is there an effect on the lifetime of pen due to the writing surface (i.e. column effect)? Conduct the appropriate test at $\alpha = 5\%$.

(d) Is there an interaction effect on the lifetime of pen due to both the brand and the writing surface? Conduct the appropriate test at $\alpha = 5\%$.

3. (Stock Index Changes 2011-2021) The data that you will use contains daily changes of six stock indices between 20112021. In this assignment, we will focus on the relationship between SP and UKX daily changes. We want to understand how the UKX daily changes (predictor) may explain the SP daily changes (response).

   (a) Using a plot show the relationship between UKX and Sp. Comment on any pattern that you may see. Does a simple linear regression model seem reasonable?

   (b) Fit and interpret the slope and intercept coefficients associated with the simple linear regression model.

   (c) Discuss the meaning of the ANOVA F-test. Obtain the R^2 value and explain what this value means.

   (d) Obtain 95% confidence intervals for the intercept and slope coefficients. Interpret your results.

   (e) Under the 0.05 significance level, test whether the slope term is significantly different from zero. Write out the null and alternative hypotheses, the test statistic value, the p-value, the decision rule and your conclusion.

   (f) Under the 0.05 significance level, test whether there is evidence to support that the slope term is significantly larger than 0.5. Write out the null and alternative hypotheses, the test statistic value, the p-value, the decision rule and your conclusion.

   (g) Give the 95% Confidence and 95% Prediction Intervals when UKX is 0. Interpret these intervals.

   (h) Look at relevant residual diagnostic plots and perform necessary test(s) to examine whether the model assumptions are satisfied. Please justify your conclusions.

4. (Properties of k_i) show that for

$$k_i = \frac{X_i - \bar{X}}{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2}$$

the following holds

(a)

$$k_1 + \ldots + k_n = 0$$

(b)

$$k_1 X_1 + \ldots + k_n X_n = 1$$

(c)

$$k_1^2 + \ldots + k_n^2 = \frac{1}{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2}$$